

Module 4

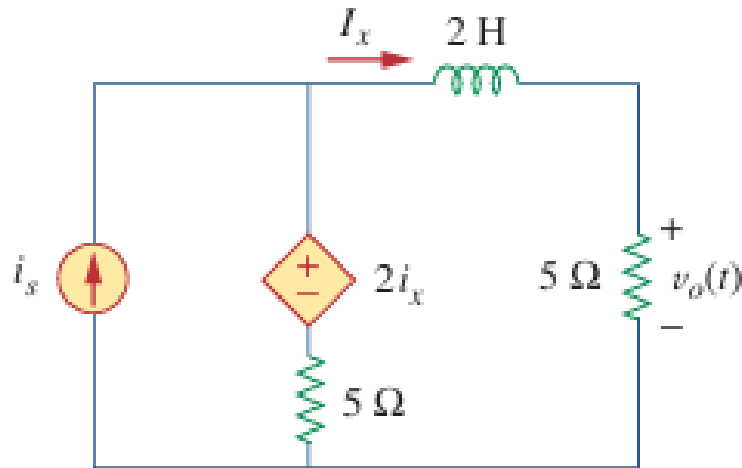
Circuit Analysis in the S Domain

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Circuit analysis using Laplace Transform

Assume that there is no initial energy stored in the circuit at $t = 0$ and that $i_s(t) = 10 u(t)$ A, a) find $V_o(s)$ using Thevenin theorem. b) apply the initial and final value theorems to find $v_o(0^-)$ and $v_o(\infty)$ and c) determine $v_o(t)$.

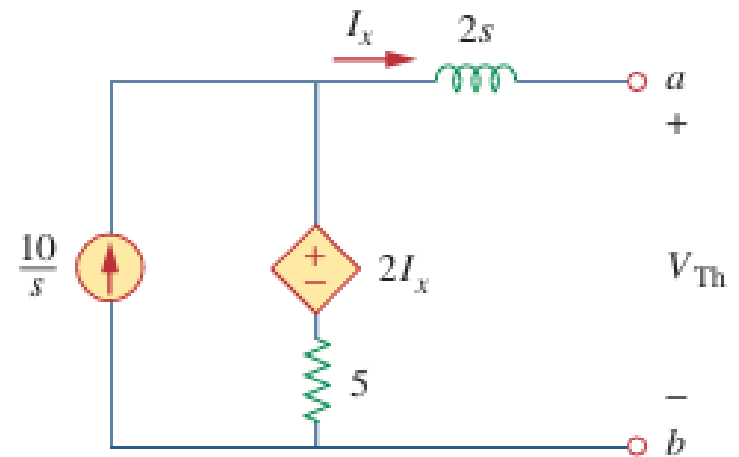


Since there is no initial energy stored in the circuit, we assume that the **initial inductor current** and **initial capacitor voltage** are zero at $t = 0$

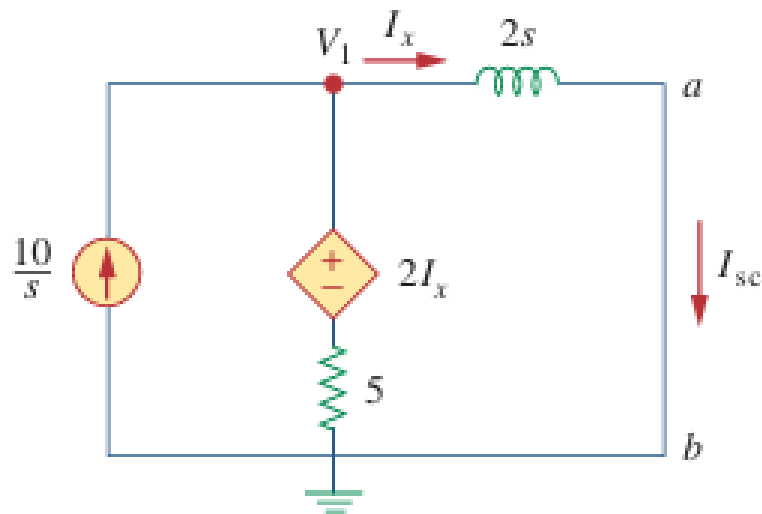
(a). Thevenin equivalent circuit: we have to find V_{oc} (V_{th}) and I_{sc}

Since $I_x = 0$

$$V_{oc} = V_{Th} = 5 \left(\frac{10}{s} \right) = \frac{50}{s}$$



Circuit analysis using Laplace Transform



nodal analysis to solve for which then leads to

$$I_{sc} (I_{sc} = I_x = V_1/2s).$$

$$-\frac{10}{s} + \frac{(V_1 - 2I_x) - 0}{5} + \frac{V_1 - 0}{2s} = 0$$

$$I_x = \frac{V_1}{2s}$$

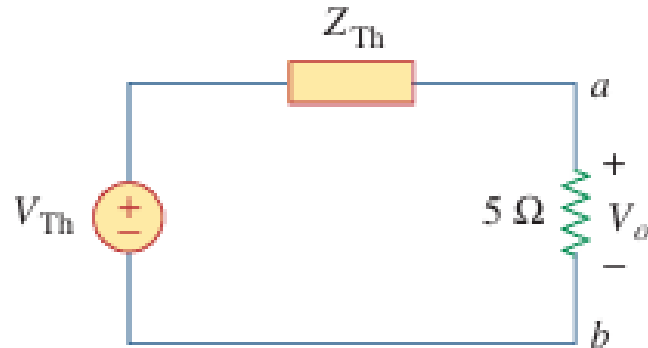
$$V_1 = \frac{100}{2s + 3}$$

$$I_{sc} = \frac{V_1}{2s} = \frac{100/(2s + 3)}{2s} = \frac{50}{s(2s + 3)}$$

$$Z_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{50/s}{50/[s(2s + 3)]} = 2s + 3$$

Circuit analysis using Laplace Transform

Thevenin equivalent at terminals a-b:



Initial value

$$f(0)$$

$$\lim_{s \rightarrow \infty} sF(s)$$

Final value

$$f(\infty)$$

$$\lim_{s \rightarrow 0} sF(s)$$

Applying voltage division rule,

$$V_o = \frac{5}{5 + Z_{Th}} V_{Th} = \frac{5}{5 + 2s + 3} \left(\frac{50}{s} \right) = \frac{250}{s(2s + 8)} = \frac{125}{s(s + 4)}$$

(b) Using the initial-value theorem and final value theorem we find:

$$v_o(0) = \lim_{s \rightarrow \infty} sV_o(s) = \lim_{s \rightarrow \infty} \frac{125}{s + 4} = \lim_{s \rightarrow \infty} \frac{125/s}{1 + 4/s} = \frac{0}{1} = 0$$

$$v_o(\infty) = \lim_{s \rightarrow 0} sV_o(s) = \lim_{s \rightarrow 0} \frac{125}{s + 4} = \frac{125}{4} = 31.25\text{ V}$$

Circuit analysis using Laplace Transform

(c) To find $v_o(t)$:

$$V_o = \frac{125}{s(s+4)} = \frac{A}{s} + \frac{B}{s+4}$$

$$A = sV_o(s) \Big|_{s=0} = \frac{125}{s+4} \Big|_{s=0} = 31.25$$

$$B = (s+4)V_o(s) \Big|_{s=-4} = \frac{125}{s} \Big|_{s=-4} = -3$$

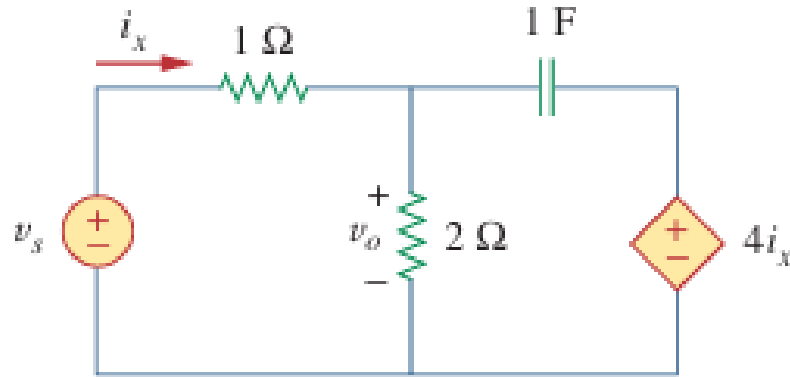
$$V_o = \frac{31.25}{s} - \frac{31.25}{s+4}$$

Taking the inverse Laplace transform gives

$$v_o(t) = 31.25(1 - e^{-4t})u(t) \text{ V}$$

Circuit analysis using Laplace Transform

1. The initial energy in the circuit below is zero at $t = 0$. Assume that $v_s(t) = 30 u(t)$ (a) Find $V_o(s)$ using the Thevenin theorem. (b) Apply the initial- and final-value theorems to find $v_o(0^-)$ and $v_o(\infty)$ and (c) Obtain $v_o(t)$.



Answer: (a) $V_o(s) = \frac{24(s+0.25)}{s(s+0.3)}$, (b) 24 V, 20 V,
(c) $(20 + 4e^{-0.3t})u(t)$ V.

Next class: Transfer function